

# Aniruddh Ballal

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Systems-oriented engineer building real-world applications grounded in underlying models—from scalable full-stack systems to physical simulation and financial modelling. Strong in end-to-end development, high-performance computing, and debugging, with a bias toward understanding how things truly behave. Currently exploring the intersection of engineering, physics, and finance.

## Education

**Manipal Institute of Technology** Manipal, KA
*B.Tech, Information Technology* *Expected May 2026*
CGPA: 8.01

**Deeksha Integrated PU College** Yelahanka, KA
Karnataka 2nd PUC (94.2%) 2022

**Daffodils English School** Sanjaynagar, KA
ICSE 10th (92.9%) 2020

## Skills

**Languages**

TypeScript, Python, Rust, JavaScript, C/C++, SQL

**Frontend**

React, Next.js, Three.js, React Native, Tailwind, GLSL

**Backend**

Node.js, Express, FastAPI, MongoDB, PostgreSQL, GridFS

**ML/Systems**

XGBoost, PuLP, SciPy, NumPy, WASM, Web Workers

**Tools**

Git, Docker, Vercel, Arduino IDE, KiCad, Postman

## Experience

**Data Analyst Intern | Qwixpert Consulting** Jan–Mar 2026
Built and deployed end-to-end full-stack internal tools, translating client data and legacy Excel workflows into scalable, high-performance solutions; diagnosed real-world problems and delivered accurate, maintainable systems.

**Frontend Web Dev Intern | Indiassetz** Jun–Jul 2025
Collaborated with design and engineering teams to enhance UI/UX using React and Material UI; implemented micro-interactions and component redesigns following best practices for clean, reusable, version-controlled code.

**Solar Physics Intern | Dept. of Physics, IISc** Jun–Aug 2024
Worked under Dr. Prantika Bhowmik on a Python-based project modelling and visualising solar magnetic field evolution and cyclical patterns using spherical harmonics, SciPy, AstroPy, NumPy, and Matplotlib; surfaced new quantitative insights into the Sun's magnetic cycle behaviour.

**Junior Avionics Member | ThrustMIT** Nov 2022–Nov 2023
Designed and built avionics systems for a student rocketry team: custom PCBs, telemetry, RF/GPS/IMU module integration with microcontrollers, control theory, embedded programming, and delivery of a hands-on rocketry workshop.

## Community

**Astronomy Club, MIT Manipal** Feb 2024 – Apr 2025
As a Managing Committee Member, completed coursework in History Philosophy of Astronomy, AstroTech, Black Holes, Astrobiology, and Data-Driven Astronomy; active role in organising club events and outreach.

**Revels PnP, MIT Manipal** Jan – Mar 2024
Led a sub-team of 4 to paint 10+ acrylic posters (14×8 each) and craft 3D models for MIT's annual cultural festival drawing 25,000+ footfalls.

**Manipal Runners' Club** Dec 2023 – Feb 2024
Co-organised the Manipal Marathon 2024 (8,000+ participants); personally coached a group of enrollees with structured training over 6 weeks leading up to the event.

## Projects — *demos & source: aniruddhballalportfolio.vercel.app*

**Sunalyse** 2025–26
3D solar coronal field viewer for 181 Carrington Rotations. Python PFSS pipeline (lmax=85, NASA/SDO FITS, haversine seeds, multiprocessed Euler); Three.js renders photosphere, per-vertex field-strength field lines, heliospheric current sheet via GLSL.

**Queen's Gambit 2.0** 2025–26
Client-side chess steganography; Rust/shakmaty compiled to WASM + Web Workers: **˜38x speedup**. AES-256-GCM via Web Crypto API with PBKDF2 key derivation.

**Credex** 2026
B2B trade finance platform (unifies VenPay, LOANlo, VenPayRL); 17 MongoDB collections, immutable audit logs. FastAPI: PuLP 90-day cash flow optimiser + XGBoost credit scorer (300–900); KYC in Backblaze B2 with signed expiring URLs.

**LOANlo** 2025
Role-based loan origination (applicant, underwriter, admin); GridFS document vault, ML credit-risk microservice, CAPTCHA + rate limiting + IP whitelisting + soft-delete governance.

**VenPay** 2025
B2B procurement: RFQ-to-PO workflow, Net30/custom deadlines, Razorpay partial payments, real-time dues tracking, post-settlement ratings. JWT auth, Cloudinary, Redux Toolkit.

**SatLies** 2026
Satellite mission simulator: RK45 orbit integrator (J2 perturbation, 3D Earth), Hohmann transfer planner (delta-V, Tsiolkovsky fuel), ground station pass predictor.